

ECON712: Midterm Exam

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There are five questions in this exam for a total of 53 points. Good Luck!!!

Question I: Linear transformation of variables (8 points)

Assume the regression model:

$$\mathbf{y} = \boldsymbol{\iota}_n \hat{\gamma} + \mathbf{X} \hat{\boldsymbol{\beta}} + \hat{\mathbf{u}}$$

where

- $\mathbf{y}$ is a $n \times 1$ vector of dependent variables
- $\boldsymbol{\iota}_n$ is an $n \times 1$ vector of ones,
- $\mathbf{X}$ is a $n \times k$ matrix of regressors,
- $\hat{\gamma}$ is a scalar OLS/MM estimate
- $\hat{\boldsymbol{\beta}}$ is a $k \times 1$ vector of OLS/MM estimates
- $\hat{\mathbf{u}}$ is a $n \times 1$ vector of residuals

Define the following transformed variables:

$$\mathbf{z} = \mathbf{y} - \boldsymbol{\iota}_n \bar{y}$$

and

$$\mathbf{W} = \mathbf{X} - \boldsymbol{\iota}_n \bar{\mathbf{x}}^T$$

with

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i \quad , \quad \bar{\mathbf{x}} = \begin{pmatrix} \bar{x}_1 \\ \bar{x}_2 \\ \vdots \\ \bar{x}_k \end{pmatrix} \quad , \quad \bar{x}_j = \frac{1}{n} \sum_{i=1}^n x_{ij}$$

Suppose that a researcher uses OLS/MM to estimate the following regression instead:

$$\mathbf{z} = \mathbf{W} \hat{\boldsymbol{\alpha}} + \hat{\boldsymbol{\epsilon}}$$

Derive the relationship between $\hat{\boldsymbol{\alpha}}$ and $\hat{\boldsymbol{\beta}}$. Are the residuals from these two regressions equal?

Question II: Finite Sample Properties of OLS (8 points)

Assume that the model $\mathbf{y} = \mathbf{X} \boldsymbol{\beta} + \mathbf{u}$ satisfies the Gauss-Markov assumptions (i.e. classical assumptions), where $\mathbf{X}$ is a $n \times k$ matrix and $\mathbf{y}$ is a $n \times 1$ vector of observations. Define a new matrix $\mathbf{Z} = a \mathbf{X}$, where $a > 0$ is a scalar and a new vector $\mathbf{s} = b \mathbf{y}$, where $b > 0$ is a scalar. Suppose a researcher draws a random sample n and run the following regression:

$$\mathbf{s} = \mathbf{Z} \hat{\boldsymbol{\beta}} + \hat{\mathbf{u}}$$

Prove whether $\hat{\boldsymbol{\beta}}$ an unbiased estimator of $\boldsymbol{\beta}$? [Hint: you need to consider two separate cases: $a \neq b$ and $a = b$]

Question III: Linear regression and selection bias

Suppose potential outcomes are defined as:

$$y_i = y_{0i} + (y_{1i} - y_{0i}) D_i,$$

where $D_i \in \{0, 1\}$ is a treatment dummy. Assume that the econometrician runs the following *population* regression (i.e. solves for β_0 and β_1 using OLS/MM):

$$y_i = \beta_0 + \beta_1 D_i + u_i$$

Using this information answer the following questions:

- (a) (5 points) Solve for β_1 in terms of conditional expectations (show your work)
- (b) (5 points) Suppose that there are homogeneous treatment effects: $y_{1i} - y_{0i} = \rho$, show that β_1 is equal to the causal effect of the treatment, defined as the potential outcome, plus a term denoting selection bias.
- (c) (5 points) Suppose now that there are heterogeneous treatment effects: $y_{1i} - y_{0i} = \rho_i$, show that β_1 is equal to the average treatment effect on all individuals, plus a term that captures the additional effect on treated individuals, plus selection bias [Hint: to answer this question you *only* need to add and subtract $\mathbb{E}[\rho_i]$, and then reorganize terms]

Question IV: Goodness of Fit (5 points)

Using projection matrices the Pythagorean theorem can be expressed as:

$$\|\mathbf{y}\|^2 = \|\mathbf{P}_X \mathbf{y}\|^2 + \|\mathbf{M}_X \mathbf{y}\|^2$$

This is the basis from which the following two versions of the R^2 are obtained:

$$R_u^2 = \frac{\|\mathbf{P}_X \mathbf{y}\|^2}{\|\mathbf{y}\|^2}, \quad R_c^2 = \frac{\|\mathbf{P}_X \mathbf{M}_t \mathbf{y}\|^2}{\|\mathbf{M}_t \mathbf{y}\|^2}$$

Assume that $\mathbf{X}$ includes the constant term. Consider the transformed variable $\mathbf{z} = \mathbf{y} - c \mathbf{1}$, prove that this transformation changes the R_u^2 , but leaves R_c^2

Question V: Conditional Expectation Function

Suppose that the random variables Y and X only take the value 0 and 1 and they have the following joint probability distribution:

	Y=0	Y=1
X=0	0.2	0.6
X=1	0.1	0.1

Using this information, answer the following questions

- (a) (7 points) Compute $\mathbb{E}[Y|X = 0]$ and $\mathbb{E}[Y|X = 1]$
- (b) (5 points) Derive the linear conditional expectation function: $m(X) = \mathbb{E}[Y|X]$
- (c) (5 points) Define $u = Y - m(X)$ and show $\mathbb{E}[u|X] = 0$. Use the LIE to show that $Cov(X, u) = 0$ [Hint: show that u is mean independent]